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Days absent from work as a result of complications associated with type 2 diabetes: Evidence from 20 years of linked national registry data in Sweden

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Abstract

Aims: To analyse days absent from work related to individual microvascular, macrovascular and other complications of type 2 diabetes (T2D) and to identify key drivers of absence.

Materials and methods: National health and socio-economic individual-level data were analysed for the years 1997 to 2016 for people with T2D, and age-, sex- and residential region-matched controls (5:1) using linkage to Swedish national administrative registers, based on personal identity numbers. Regression analyses accounting for individual-level clustering and education were estimated to obtain days absent by individual complications. Alternative analyses, for example, workforce indicator and age subgroups, were explored for robustness and comparison purposes.

Results: A total of 413 000 people with T2D aged <66 years, comprising 4.9 million person-years, was included. The crude proportion with any absence was higher among those with T2D compared to controls (47% vs. 26%) in the index year, and the median (IQR) number of days was higher (223 [77;359] vs. 196 [59;352]) if any absence. Regression analyses showed that complications per se were a key driver of days absent: stroke (+102 days); end-stage renal disease (+70 days); severe vision loss (+56 days); and angina pectoris, heart failure, and osteoarthritis (+53 days each). The alternative analyses showed similar levels of days absent and age subgroups differed in expected directions.

Conclusions: This study provides evidence of the persisting impact on productivity from complications that supports continued efforts to reduce risk factors in T2D. Future studies on burden of disease and economic evaluations of new therapies and disease management may use this new set of complication-specific estimates to improve understanding of the value of reducing complications.

S.P. and P.J. should be considered joint first authors.

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KEYWORDS

costs and cost analysis, days absent from work, diabetes complications, diabetes mellitus, disability, insurance, longitudinal register data, sick leave, type 2

1 | INTRODUCTION

Type 2 diabetes (T2D) is a prevalent and costly chronic disease associated with several short- and long-term health conditions, including macrovascular and microvascular complications. Awareness of T2D and its consequences has risen over the past decades following the publication of the UK Prospective Diabetes Study.^{1,2} Guidelines for diabetes management^{3,4} suggest further personalization of treatment strategies and targeted use of new treatment technologies (glucose-lowering medications, blood glucose monitoring aids, etc.) as tools for improving outcomes and delaying onset of complications.

The burden of complications has been highlighted as a key driver of the total costs related to T2D, and is known to stretch beyond the health sector.^{5,6,15}

Costs related to reduced productivity, including, for example, workdays absent, reduced labour force participation and premature mortality, constitute a substantial part of the cost burden.^{5,6} The American Diabetes Association (ADA) estimated that 28% of total costs (\$90 of \$327bn) were attributable to reduced productivity in 2017.⁵ A higher proportion of the total cost of T2D was attributed to reduced productivity (£13bn; 60%) in 2010/2011 in the United Kingdom.⁶ Nevertheless, two reviews highlights the paucity of solid and up-to-date results on productivity loss in diabetes properly handling confounding and possible endogeneity.^{7,8} The consequences of T2D complications for work capacity are context-specific; absenteeism, presenteeism and unemployment may be associated with overall labour-force participation and social insurance coverage in a given country.

Many T2D complications occur already in those in age groups active in the labour market.⁹ However, evidence of the size of productivity loss caused by individual complications is limited, and issues of key drivers of absence from work are rarely addressed. To date, most studies have been based on less robust analyses (eg, cross-sectional surveys with self-reporting and recall bias⁵; short observation spans in observational data;⁶ or conducted in limited sample sizes).⁷ Using large, representative samples is one remedy for several challenges associated with analyses in this area. A Danish register study analysed days absent from work as a result of 10 diabetes-related complications in a total of 15 000 people in the labour force with diabetes (type 1 [T1D] and T2D)) in 1997 to 2002. They found that people with, for example, myocardial infarction (MI), heart failure and renal disease had approximately 100 days absent from work, of which 50 to 80 were attributable to the complication per se.¹⁰ Moreover, the reduced work capacity remained at similar levels in subsequent years.

Information on the true consequences of T2D complications for work capacity is of importance to health sector decision-makers at all levels, from policy makers deciding on guidelines, to physicians and nurses meeting people with T2D. It is also essential for health-

economic evaluations of new therapies and disease management programmes to have estimates of production loss by individual complications to estimate accurately the value of improving risk factor control.

The aim of the present register-based study was to analyse days absent from work related to a broad set of individual microvascular, macrovascular and other complications of T2D in Sweden.

2 | MATERIALS AND METHODS

2.1 | Study design

In the present study, we analysed a 20-year-long panel of retrospective cross-linked individual-level health, social insurance, and socio-economic data from Sweden. The National Board of Health and Welfare identified people with diabetes and provided data from the National Patient Register (NPR), on outpatient and inpatient hospital healthcare,¹¹ and from the National Prescribed Drugs Register (NPDR) on filled prescriptions of selected drugs.¹² Presence of a complication ("exposure") and its first observation was derived using main diagnoses from hospital admissions and physician visits in the NPR (Table A1).

Days absent from work ("outcome") were obtained from *Försäkringskassan*, the Swedish Social Insurance Agency, containing start and stop dates for all periods with sick pay, including short-term sickness and rehabilitation benefits (covering transient reduction in work capacity), and long-term sickness and activity compensation (covering an expected permanent inability to work and with a lower compensation rate than sickness and rehabilitation benefits). Periods of absence <14 days are not registered at *Försäkringskassan*, but for all registered periods of absence, the total number of days are included. Control subjects, to be matched to people with diabetes (5:1), were randomly selected from the Register of the Total Population with the same year of birth, sex and residential region at index date by Statistics Sweden. Socio-economic data from the Longitudinal Integrated Database for Health Insurance and Labour Market Studies (LISA) were added.¹³ The strengths and limitations of data sources as pertinent to this study are described in the Supporting Information (Table S1). The linkage of data using personal identification numbers in multiple national registers was approved by the Ethical Review Board, Lund (dnr 2018-349) and the Swedish Ethical Review Authority (dnr 2019-01260).

2.2 | Population

The full study database included people in age groups active in the labour market (16–70 years) that at any point during

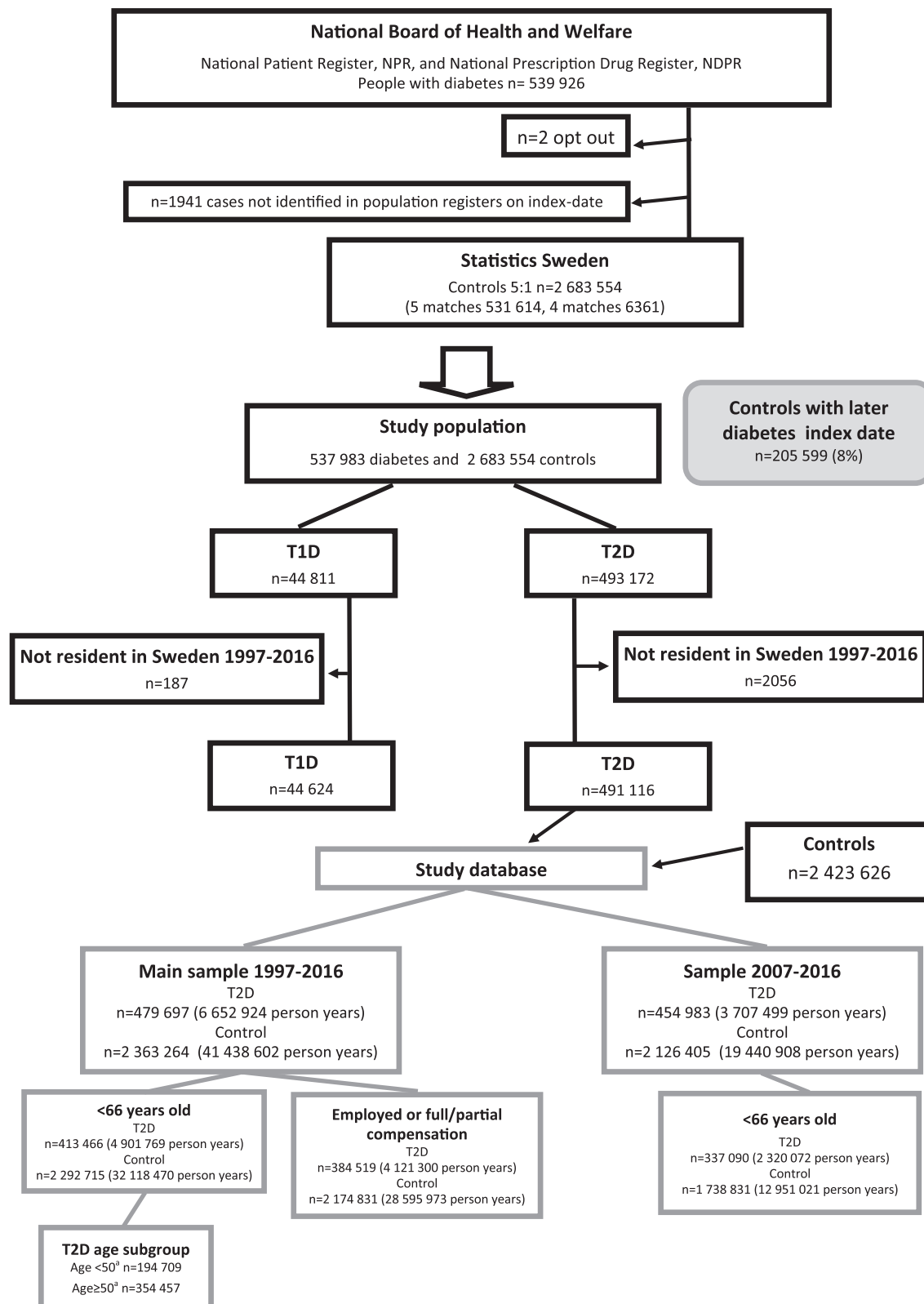


FIGURE 1 Population flow chart (black boxes) and subsets for main, sensitivity and subgroup analyses (blue boxes). T1D, type 1 diabetes; T2D, type 2 diabetes

1997–2016 fulfilled at least one of the following criteria: (1) at least one hospital visit or inpatient admission recorded in the NPR 1997 to 2016 with diabetes as the main or secondary diagnosis according to the International Classification of Diseases

version 10 (ICD-10) codes (T1D [E10], T2D [E11] or unspecified diabetes [E14]); or (2) at least two dispensed prescriptions of glucose-lowering medication (Anatomic Therapeutic Chemical classification system [ATC] codes, A10) with dispense dates not

without days absent from work, and with and without complications. The second term, $\beta_{(T2D)}$, is the basic T2D effect not attributed to a specific complication and the basic difference between T2D and controls. The primary interest for this study was the days attributed to each complication, defined as the sum of the third and fourth terms $\beta_{k(Comp\ k)} + \beta_{k+16(T2D \times Comp\ k)}$. This sum represents the estimated average additional number of days absent in the estimation sample considering its distribution of age, sex and other characteristics.

2.5 | Statistical analyses

Statistical analyses were conducted in statistical package STATA version 15.²² Demographic and socio-economic characteristics in the index year for T2D and controls were compared using ordinary descriptive statistics and standardized difference from the `stddiff` module. The presence of each complication was described using two measures: proportion of individuals with the complication at least once per 100 000 individuals ("ever had"), and rate per 100 000 person-years ("long-term burden of living with complications").

Ordinary least squares regression analyses with individual-level clustered standard errors were used to estimate days absent for people with T2D and controls, controlling for level of education, in the longitudinal panel. Regression analysis was considered suitable as it accounts for possible presence of multiple conditions. The estimated coefficients represent the marginal contribution of each complication to the total number of days absent. Large samples allow use of simple methods when the underlying distribution of the dependent variable is non-normal as the analysis depends on sample means and variances.²³

2.6 | Ethics

The linkage of individual-level longitudinal data from multiple national registers was approved by the Ethical Review Board (Regionala etikprövningsnämnden Lund dnr. 2018-349; Etikprövningsmyndigheten dnr 2019-01260). In accordance with approval, information about the study along with information on the opt-out process was published in a national newspaper and posted at the Swedish Institute for Health Economics website before initiating the process identification of the study population at the National Board of Health and Welfare and at Statistics Sweden. Two persons opted out from the study.

3 | RESULTS

3.1 | Population characteristics

Table 1 shows the demographic and socio-economic characteristics as measured in the index year for the main sample. The absolute standardized difference was <0.10 for the matching variables “year of birth” and “sex”, while “education” had an absolute standardized difference of >0.10 , indicating an imbalance between the groups which

was accounted for in the regression analyses. The overall employment rate was lower in the T2D group (59% vs. 73% for controls). A larger proportion of the T2D group had days absent from work with sickness and activity compensation (47% vs. 26%) and among them, people with T2D had more days with compensation (100 vs. 52 days).

People with T2D in age groups active in the labour market (ages <66 years) were more likely to have complications compared to controls, as shown by the proportion of individuals with at least one indication during the study period (Table 2). The two most common complications were eye disease (27%) and kidney disease (13%). Three heart conditions were also markedly higher among people with T2D compared to controls: ischaemic heart disease (5.8% vs. 2.0%), angina pectoris (7.8% vs. 2.9) and MI (6.8% vs. 2.2%).

The presence of complications was low, generally below 1000 per 100 000 person-years, in this population-based sample that also covered early years with T2D (Table 2). However, there was a correspondence between complications showing a higher rate of person-years and the proportion ever having the complication.

3.2 | Days absent from work attributable to diabetes complications

People with T2D and any of the listed complications can be expected to be absent from work for 2 to 3 months per year (Figure 2). A substantial proportion was attributed to the complication (darker section of the bars in Figure 2). End-stage renal disease (ESRD) and stroke were associated with the highest absence (141 and 174 days, respectively, for people with T2D). On an annual basis, 70 days (49%) were directly attributed to ESRD and 102 (59%) to stroke state. However, new stroke events were not associated with an increase in absenteeism, as indicated by a mean (SE) of 18.0 (1.3) days of absence (Table 3), which potentially reflects the already reduced work capacity resulting from first stroke or other comorbidities.

The group of complications categorized as “Other complications or events”, was also associated with substantial absence, eg, hypoglycaemia^a and osteoarthritis were associated with 90 and 125 days, respectively.

There was a general correspondence between high total number of days and high number of days attributed to the complication for people with T2D. For controls, this pattern had some notable exceptions, including amputation and peripheral vascular disease. People with T2D were more likely to have had an amputation (T2D 780 per 100 000; control 22 per 100 000), and it is possible that amputations among controls, if they occurred, were of greater severity because of underlying causes such as sepsis or injuries.

People without any complications, or diabetes diagnosis, had on average 42 days absent from work (Supporting Information Table S7). This reflected the age structure generated by the study design based on people with T2D: 52% of observed person-years fell within the age group 50 to 65 years, and 48% within the age group <50 years. The average, non-complication-specific, diabetes effect was 29 days per year.

TABLE 1 Demographic and socio-economic description: Measurement in index year^a

Demographic characteristics and education	T2D, N = 458 938	Controls, N = 2 250 512	Std diff.
Year of birth ^b			
Before 1940	80 092 (17)	393 934 (17)	0.007
1940–1949	169 480 (37)	824 266 (37)	
1950–1959	114 641 (25)	565 473 (25)	
1960–1969	60 247 (13)	297 854 (13)	
1970–1979	23 439 (5)	115 109 (5)	
1980–1989	8818 (2)	42 949 (2)	
During or after 1990	2221 (0.5)	10 927 (0.5)	
Age	56.6 (10.7)	56.6 (10.7)	−0.001
Sex ^b			
Men	276 252 (60)	1 347 268 (60)	0.007
Women	182 686 (40)	903 244 (40)	
Marital status ^c			
Married/cohabiting	260 342 (58)	1 386 148 (62)	0.132
Single with/without children	191 823 (42)	852 950 (38)	
Missing or unclear information	6773 (1.5)	11 414 (0.5)	
Education ^c			
Compulsory	153 342 (33)	596 968 (27)	0.228
Upper secondary	207 163 (45)	978 043 (44)	
University	92 331 (20)	655 762 (29)	
Missing	6102 (1)	19 739 (1)	
Labour market active, ages <66 years	T2D, N = 352 607	Controls, N = 1 741 265	Std diff.
Socio-economic characteristics			
Employed in November	208 563 (59)	1 264 059 (73)	0.286
Studying	6960 (2.0)	38 453 (2.2)	0.016
Parental leave	21 783 (6.2)	150 625 (8.7)	0.094
Job seeking	13 445 (3.8)	47 156 (2.7)	−0.062
Military service	17 (<0.1)	419 (<0.1)	0.016
No registration of workplace	118 757 (34)	367 824 (21)	−0.284
Sickness and activity compensation ^d	91 446 (26)	251 092 (14)	−0.290
Sickness benefits only ^b	74 665 (29)	197 038 (13)	−0.385
Unemployment	25 624 (7.3)	122 557 (7.0)	−0.008
Days of unemployment, mean (SD)	14.4 (55.2)	11.4 (49.0)	−0.057
Days of unemployment >0, mean (SD)	132 (114)	124 (113)	−0.077
Days of unemployment >0, median (IQR)	112 (28; 214)	99 (14; 203)	<0.001 ^f
Absent from work with sickness benefits or sickness and activity compensation at least once	169 261 (47)	453 251 (26)	−0.459
Days absent; mean (SD)	100 (140)	52 (111)	−0.384
Sample with at least 1 day of absence	T2D N = 169 261	Control N = 453 251	Std diff
All			
Days, mean (SD)	210 (134)	198 (135)	−0.0887
Days, median (IQR)	223 (77; 359)	196 (59; 352)	<0.001 ^f
People with sickness benefit only and at least 1 day absent from work ^d			
Days, mean (SD)	116 (116)	99 (110)	−0.149
Days, median (IQR)	67 (21; 188)	47 (14; 152)	<0.001 ^f

(Continues)

TABLE 1 (Continued)

Demographic characteristics and education	T2D, N = 458 938	Controls, N = 2 250 512	Std diff.
People with sickness and activity compensation and at least 1 day absent from work			
Days, mean (SD)	290 (89)	278 (96)	−0.134
Days, median (IQR)	333 (222; 365)	314 (190; 365)	<0.001 ^f

Abbreviations: IQR, interquartile range; Std diff., standardized difference; T2D, type 2 diabetes.

Number of observations (%) unless otherwise stated.

^an = 2335 (T2D, n = 394; controls, n = 1941) had index year <1997 based on hospital admission date and n = 69 920 individuals (T2D, n = 11 805; controls, n = 58 115) had index date in 2017 or 2018. These individuals were not included because of lack of data from index year in main sample.

^bMatching variable.

^cMissing data for 6771 individuals (1.5%) in T2D group and 11 412 (0.5%) in the control group in Longitudinal Integrated Database for Health Insurance and Labour Market Studies (LISA) in index year.

^dRegistration of social insurance payments from sickness and activity compensation in year.

^eDays from sickness benefits only and no registration in year of sickness and activity compensation.

^fP value Wilcoxon rank-sum test.

TABLE 2 Diabetes complications in the period 1997 to 2016 for people with type 2 diabetes (n = 413 466) and controls (n = 2 292 715) in main analysis (sample age < 66 years)

Diabetes complication or concomitant condition	Proportion per 100 000 individuals				Rate per 100 000 person-years			
	T2D	Control	P	Std diff. ^a	T2D	Control	P	Std diff. ^a
Microvascular complications								
Eye disease and diabetic retinopathy	27 085	15 193	<0.001	0.294	6404	1909	<0.001	0.277
Vision loss or blindness	235	116	<0.001	0.029	23	9	<0.001	0.024
Lower extremity disease								
Neuropathy	4191	2167	<0.001	0.116	510	212	<0.001	0.108
Peripheral vascular disease	3517	446	<0.001	0.222	580	52	<0.001	0.157
Amputation	780	22	<0.001	0.120	83	2	<0.001	0.074
Kidney disease	13 471	1437	<0.001	0.471	3233	200	<0.001	0.360
ESRD ^b	1523	273	<0.001	0.133	586	83	<0.001	0.097
Macrovascular complications								
Ischaemic heart disease	5874	1955	<0.001	0.203	863	231	<0.001	0.158
Angina pectoris	7774	2918	<0.001	0.217	1061	300	<.001	0.195
Acute MI	6786	2213	<0.001	0.222	583	170	<0.001	0.162
Stroke	4450	1613	<0.001	0.166	398	128	<0.001	0.116
Heart failure	3707	893	<0.001	0.189	556	105	<0.001	0.126
Atrial fibrillation	3710	2352	<0.001	0.079	634	389	<0.001	0.068
Other sudden death, cause unknown	5	4	0.324	0.002	NA	NA		
Other complications and events								
Hypoglycaemia	2499	20	<0.001	0.224	331			
Hyperglycaemia	1873	8	<0.001	0.194	186			
Ketoacidosis	673	0	<0.001	0.116	77			
Coma	1231	65	<.001	0.146	124			
Diabetic foot including ulcer	3443	928	<.001	0.173	447			
Osteoarthritis	1710	186	<.001	0.158	171	15	<0.001	0.097

Abbreviations: ESRD, end-stage renal disease; MI, myocardial infarction.

Number of unique individuals with complications per 100 000 in population (proportion), and number of person-years with complications per 100 000 person-years (rate).

^aAbsolute standardized difference.

^bESRD with dialysis or kidney transplantation.

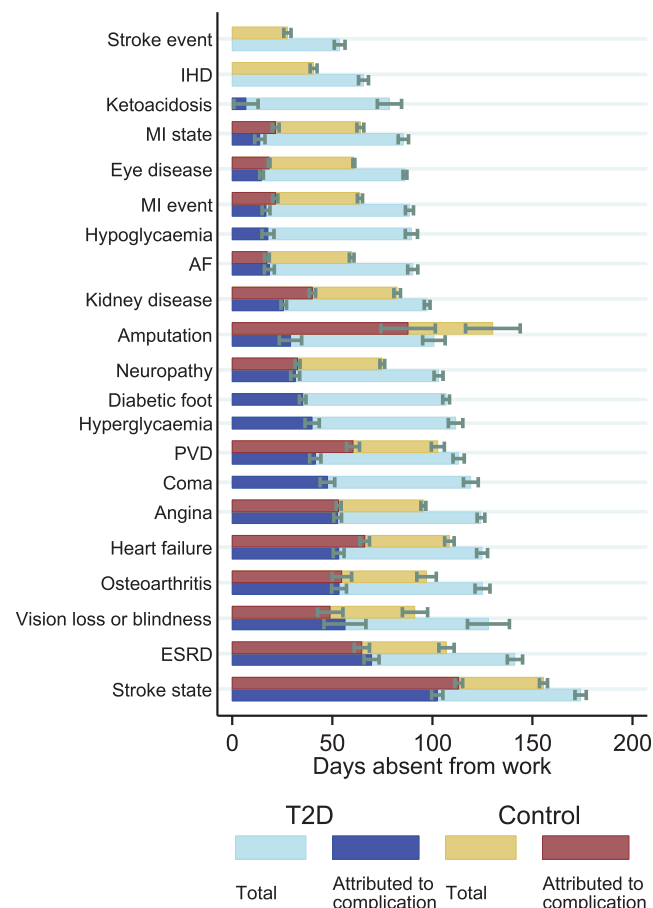


FIGURE 2 Total days absent from work for type 2 diabetes (T2D; light-blue) and controls (yellow), by individual complication and proportion of days that were attributed to the complication per se (T2D, blue; controls, red). Sorted from lowest total number of days per person with the complication to highest number of annual days. Error bars show 95% confidence intervals from robust SEM estimates. AF, atrial fibrillation; ESRD, end-stage renal disease with dialysis or kidney transplantation; IHD, ischaemic heart disease; MI, acute myocardial infarction; PVD, peripheral vascular disease

3.3 | Sensitivity and subgroup analyses

Table 3 shows the annual days absent per individual with T2D attributed to each complication from four sensitivity and subgroup analyses. The sensitivity analyses showed that people employed or with sickness/activity compensation differed slightly, with both fewer days (e.g. atrial fibrillation and osteoarthritis) and more days (e.g., vision loss or blindness, and heart failure) attributed to T2D complications, compared to people included in the main analysis. However, both analyses confirmed that days absent because of complications constituted an important component of the burden of disease and the slight differences were expected due to different sample definitions.

The subgroup analysis including years 2007 to 2016, showed a lower number of days absent attributed to each complication compared to the main analyses. The largest difference was estimated for angina pectoris (38.9 vs. 52.6 days). This finding was expected

because of the inclusion of less severe cases of T2D identified through the NPDR.

The subgroup analyses of those aged 16–49 years and those aged 50 to 65 years showed that, although the number of total days absent from work was higher in the older group (Supporting Information Table S8), the number of days attributed to each complication was generally greater in the younger age group. Sensitivity and subgroup analyses are further commented on in the Supporting Information pp. 25–34.

4 | DISCUSSION

The present study shows the substantial and persisting impact on the number of days absent from work attributable to T2D complications. Stroke, ESRD and vision loss or blindness were associated with more than 3 months absence, of which approximately half was attributed to the complication per se. The results also confirm that complications are an important underlying cause of the higher absenteeism observed among people with T2D compared to controls. For example, an MI in a 50-year-old person with T2D adds approximately 4 weeks of absence in the year of the event but will, all else being equal, add more than 1 year of absence before retirement at the age of 65 years. Similarly, kidney disease was associated with an average of 26 days absent each year. Approximately 13% of all individuals with T2D in the main analysis had kidney disease and it was observed in 3% of person-years. Onset of kidney disease in a 50-year-old would then correspond to more than 1 full year of absence (385 days) before retirement. Our results indicate that complications per se had a similar impact on people with T2D and controls, that is, having a stroke is, on average, as bad regardless if the person was in the T2D or in the control group in terms of work capacity. Nonetheless, we found that complications were more common among people with T2D (Table 2) and the results show that a large proportion of the total days absent from work was also associated with T2D in itself.

These results may be used in analyses estimating the burden of diabetes complications, together with information on expected number of people with each complication. The data can also be used in health-economic models to estimate indirect costs, for example, in long-term cost-effectiveness analyses to capture the broader societal perspective. The Supporting Information provides a practical guide on how to use the study results in economic models (pp. 18–24).

To our knowledge this is the first comprehensive estimation of days absent attributed to the T2D complications usually included in health-economic evaluations. The previous Danish study¹¹ included 15 000 individuals with diabetes (T1D and T2D) in the work force using data from 1997 to 2002. Our data cover more than 400 000 people with T2D and use a longer and more recent panel, 1997 to 2016, enabling extensive sensitivity and subgroup analyses. We also distinguished between types of diabetes to capture days absent associated with complications in people with T2D and to analyse differences

TABLE 3 Days absent from work per year per individual attributed to diabetes complications for people with type 2 diabetes

Diabetes complications ^a	Main analysis		Sensitivity analysis		Subsamples					
	Years 1997–2016, ages <66 years		Employed or with compensation ^e		Years 2007–2016, ages <66 years		Ages 16–49 years		Ages 50–65 years	
	T2D (n = 413 466)		T2D (n = 384 519)		T2D (n = 337 090)		T2D (n = 194 709)		T2D (n = 354 457)	
	Controls (n = 2 292 715)		Controls (n = 2 174 831)		Controls (n = 1 591 737)		Controls (n = 1 248 922)		Controls (n = 1 996 611)	
	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Microvascular complication										
Eye disease and diabetic retinopathy ^b	14.6	0.5	10.4	0.5	15.6	0.5	11.5	0.8	8.6	0.5
Vision loss or blindness	56.3	5.3	70.7	5.7	62.0	5.3	67.1	9.9	51.8	5.7
Lower extremity disease										
Neuropathy	31.4	1.1	29.9	1.2	32.4	1.1	33.7	1.9	27.3	1.3
Peripheral vascular disease	41.4	1.4	41.0	1.5	33.7	1.6	43.1	3.5	37.1	1.5
Amputation	29.1	2.9	35.9	3.2	21.9	3.3	44.0	7.5	27.9	2.9
Kidney disease	25.7	0.7	27.0	0.8	21.1	0.8	26.9	1.2	22.9	0.8
ESRD ^c	69.6	1.9	73.5	2.1	67.8	2.1	88.4	3.2	62.7	2.2
Macrovascular complication										
Ischaemic heart disease	−6.1	1.2	−10.8	1.3	0.2	1.4	−4.8	3.6	−8.1	1.3
Angina pectoris	52.6	1.0	50.5	1.0	38.9	1.1	61.2	2.7	43.4	1.0
Acute MI										
Event	16.9	1.0	25.7	1.1	11.9	1.3	20.0	2.7	15.5	1.1
State	13.8	1.3	11.4	1.4	8.1	1.4	22.9	3.6	7.3	1.3
Stroke										
Event	−18.0	1.3	−5.0	1.4	−19.5	1.8	−11.4	4.3	−19.6	1.4
State	102.4	1.4	97.3	1.5	96.3	1.6	113.5	4.6	92.3	1.4
Heart failure	53.2	1.4	58.0	1.5	49.2	1.6	65.9	3.9	47.9	1.4
Atrial fibrillation	18.5	1.3	6.4	1.3	11.9	1.3	19.3	3.7	10.1	1.3
Other complication or event										
Hypoglycaemia ^d	17.9	1.5	20.7	1.7	16.4	1.7	15.7	2.4	24.1	1.9
Hyperglycaemia ^d	39.9	1.8	56.5	2.0	39.3	2.0	38.8	2.9	44.3	2.2
Ketoacidosis ^d	6.9	3.1	29.2	3.6	13.7	3.2	8.2	3.8	28.0	4.7
Coma ^d	47.6	1.9	57.1	2.1	40.2	2.9	42.9	3.0	51.8	2.3
Diabetic foot or ulcer ^d	35.2	0.8	35.3	0.9	35.4	0.8	40.1	1.5	31.9	0.9
Osteoarthritis	53.4	1.9	34.1	1.9	47.8	1.9	62.4	8.0	42.8	1.9

Abbreviations: ESRD, end-stage renal disease; MI, myocardial infarction.

Calculated from coefficients in regression analysis controlling for educational level (Supporting Information Tables S7 and S8).

^aDiabetes complication refers to all diagnoses conditions and procedures analysed in the study. The conditions may occur in people with and without diabetes.

^bNot vision loss and blindness.^cESRD with dialysis or kidney transplantation.

^dDiabetes-specific complication with low/no frequency among controls; no interaction effect estimated.

^eIncludes people that are employed or on full/partial sickness and activity compensation.

in impact compared to controls. For some complications, the results from our study are in line with the Danish study (eg, for angina pectoris with 53 vs. 55 days absent). For other complications, we found fewer days absent (eg, peripheral vascular disease 41 vs. 65 days, amputation 29 vs. 104 days). While research designs and country settings differ in important aspects, this disparity was expected in light of changes in diabetes management over the past 20 years.

Previous reviews^{7,8} and burden-of-disease studies^{5,6} have pointed at a handful of days of absenteeism attributed to T2D. Many previous studies were based on limited sample surveys or observational data and did not distinguish complications as a driver of absence. Our national longitudinal data highlight T2D complications as a key driver of absence, and postponing onset of complications will have immediate benefits also in terms of retained work capacity. The

comparatively high basic T2D effect (42 days) is partly explained by the fact that our data cover sickness and activity compensation (early retirement for health causes) based on the structure of the Swedish social insurance system. Our estimates should not be compared with the short-term sick-leave reported in studies such as the US burden of diabetes,⁵ but with the combination of workdays absent and reduced labour force participation as a result of disability.

An important strength of the present study lies in its design, which avoided double-counting of days absent for individual complications. The regression analyses distributed up to 365 calendar sick days across complications and implicitly accounted for multiple complications. Administrative data sources also avoided recollection bias where complications were defined by main diagnosis in the NPR and days absent were based on actual payments registered at *Försäkringskassan*.

Our study design using populations controls allowed us to explore if complications, such as cardiovascular conditions, generated more days absent in people with T2D, which was not addressed previously,¹¹ but we found no strong evidence that this was the case. Moreover, the administrative data sources allowed controls to switch later to the T2D group. This dynamic approach was a preferred strategy to avoid bias, as T2D is a common disease, and we avoided conditioning the population on information about future events.

The NDR^b reported approximately 131 000 individuals with T2D (ages 18–65 years) in 2016,²⁴ which suggests that our study population with approximately 199 000 individuals in 2016 represents the majority of people with T2D in Sweden in labour market age groups.

The study has some limitations. Sick periods <14 days are not registered at *Försäkringskassan* and people with no registered days in our data may have multiple short sick periods. In such instances, our results are then biased downwards if T2D complications were causing these sick periods.

Presence of diabetes complications was sourced from 20 years of individual-level NPR data, capturing complications serious enough to require outpatient physician care or inpatient admission. Thus, early stages of complications and their impact on days absent is included in the basic diabetes effect. The same is true for our event-based complications and, for example, milder hypoglycaemic events not associated with hospital care.

Production loss is, by nature, setting-dependent, reflecting current labour market and social insurance institutions. Our results represent those of working age with T2D in Sweden, which may differ from other country settings. Overall and age-specific eligibility and compensation levels and duration may be high in Sweden compared to other countries. If this is the case, estimates from this study may be higher than elsewhere. However, we had access to complete reporting on long-term sickness and activity compensation payment for 20 years. Data on long-term absence in other studies often rely on surveys using self-reports on presence of complications and on productivity loss and are vulnerable to recall bias.^{5,7,8}

For society, productivity losses associated with complications of T2D are substantial: more than 50 days per year per person were attributed to the onset of vision loss or blindness, ESRD, angina

pectoris, stroke, heart failure and osteoarthritis in T2D. As expected, people with T2D were more likely to have complications and these generate absenteeism. However, there is no firm evidence that when a complication occurs, people with T2D are more absent from work than controls with the same complication. Treatments which can prevent or delay the onset of, for example, cardiovascular disease play an important role in reducing the economic burden on society. Two new classes of glucose-lowering agents, glucagon-like peptide-1 receptor agonists and sodium-glucose co-transporter-2 inhibitors, are highlighted in joint guidelines from the ADA and the European Association for the Study of Diabetes for providing health and quality-of-life benefits for people with T2D.^{24,25}

CONFLICTS OF INTEREST

S.P., E.A., P.L. and K.S.C. are employees of the Swedish Institute for Health Economics, IHE, Sweden, a consultancy that received funding for the study from Novo Nordisk A/S, Copenhagen, Denmark. P.J., D.T. and B.L.T. are employees of Novo Nordisk A/S, Søborg, Denmark. P.J. and B.L.T. are shareholders of Novo Nordisk A/S. J.J. has received research grants from Novo Nordisk and Medtronic, speaker honoraria from Abbott, AstraZeneca, Boehringer Ingelheim, Eli Lilly, MSD, Nordic Infucare, Novo Nordisk, Medtronic and Sanofi, and has served as a member of advisory boards for Abbott, AstraZeneca, Boehringer Ingelheim, Eli Lilly, Novo Nordisk, Medtronic and Nordic Infucare.

AUTHOR CONTRIBUTIONS

All authors confirm that they meet the International Committee of Medical Journal Editors uniform requirements for authorship and that they have contributed to: design and collection of data; or critical analysis and interpretation of the data; and drafting/critically revising the article. All authors share the final responsibility for the content of the manuscript and the decision to submit it for publication. Material preparation, data collection and analysis were performed by S.P., K.S.C. and E.A. The first draft of the manuscript was written by S.P. and K.S.C.

DATA ACCESSIBILITY

Subject-level data from national registers hosted by Swedish national authorities are available for research after formal evaluation of the research protocol by the Swedish Ethical Review Authority and by the respective national authorities providing data. Permission to conduct research is granted on a case by case basis to a limited number of named persons.

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ENDNOTES

^aSevere hypoglycaemia for which hospital-based care was needed.

^bNDR was reported to cover about 92% of the Swedish T2D population in 2016 [24].

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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APPENDIX

Diabetes complications: Empirical definitions.

TABLE A1 Codes for complications of diabetes and concomitant disease for registry-based identification using ICD 10 codes and hospital procedure codes (KVA)

Conditions and events (abbreviation in text)	Comments	ICD 10 codes or procedural codes	Reference
Angina pectoris (AP)		I20	17, 20, 26
Acute MI		I21	17, 20, 21
Ischaemic heart disease		I22-I25	17, 20, 21
Atrial fibrillation		I48	21, 26, 27
Heart failure		I50	17, 20, 21, 26, 27
Stroke		I61, I63, I64, I67.9	17, 20, 26, 27
Peripheral vascular disease		I70.2, I73.1, I73.9, I79.2, E11.5, E14.5	17, 26
Lower extremity amputation		KVÅ: NEQ19, NEQ99, NFAQ09, NFAQ19, NFAQ99, NGQ09, NGQ19, NGQ99, NHQ09, NHQ11, NHQ12, NHQ13, NHQ14, NHQ16, NHQ17, NHQ99	10, 17, 26
Diabetic foot or ulcer		E11.6, E14.4, L00, L03, L08, L97	10
Eye disease and diabetic retinopathy	Not vision loss and blindness	H0-H4, H50-H53, H55, H57-H59, E103, E113, E143, KVÅ: DT006	17
Vision loss or blindness	Severe vision loss /blindness in one or two eyes	H54	17
Neuropathy	Symptomatic neuropathy	E11.4D, G62.9, G56, G58.9, G99, G63.2, G73.3, M14.6	10
Kidney disease	Any stage	E11.2, E14.2, E11.7, E11.8, E14.7, E14.8, R809, N00-N08, N10-N19, N28.9	17, 27
ESRD	ESRD actively treated with dialysis or kidney transplantation	Dialysis Z49, Z94, Z99.2 KVÅ dialysis DR015, DR016, DR017 KVÅ transplantation KAS10, KAS20, VF420	17, 27
Hyperglycaemia		R73.9	18, 26
Ketoacidosis	With/without coma	E11.0A, E11.1A, E14.1	18, 26
Hypoglycaemia	Without/with coma	E11.0C, E11.6, E11.6A, E14.0C, E14.6A, E16.0, E16.1, E16.2	18, 26
Coma	Unspecified	E11.0, E14.0, R40.2	18, 26
Osteoarthritis		M15-M19	

Abbreviations: ESRD, end-stage renal disease; KVÅ, *Klassifikation av vårdåtgärder* [Classification of healthcare interventions; National Board of Health and Welfare]; MI, myocardial infarction.